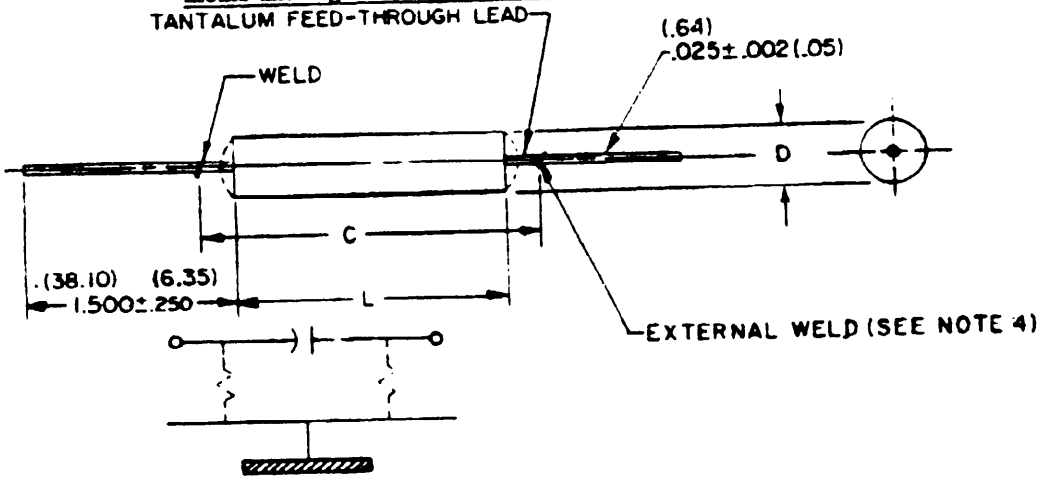
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (NONPOLARIZED, HIGH ETCHED FOIL) - 85°C (VOLTAGE DERATED TO 125°C), STYLES CL72 AND CL73

INACTIVE FOR NEW EQUIPMENT DESIGN AFTER DATE OF THIS SPECIFICATION; USE MIL-C-39006/17

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-3965.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



CIRCUIT DIAGRAM

Case size	Dimensions			
	1/ L ±.062 (1.57)	D ±.016 (.41)	D Max	C Max
		CL72	CL73	
E1	.688 (17.48)	.188 (4.78)	.219 (5.56)	1.188 (30.18)
E2	.906 (23.01)	.281 (7.14)	.312 (7.92)	1.406 (35.71)
E3	1.438 (36.53)	.375 (9.53)	.406 (10.31)	1.938 (49.23)
E4	2.125 (53.98)	.375 (9.53)	.406 (10.31)	2.625 (66.68)
E5	2.750 (69.85)	.375 (9.53)	.406 (10.31)	3.250 (82.55)

1/ For style CL73, nonshrinkable sleeving shall extend .016 inch minimum, .062 inch maximum, beyond each end of the case; shrinkable sleeving shall lap over the ends of the case.

NOTES:
1. All dimensions in inches.
2. Millimeters are in parentheses.
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
4. The weld shall not be enclosed in the end seal.

FIGURE 1. Styles CL72 and CL73 capacitors.

FSC 5910

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TABLE I. Capacitor characteristics.

Type designation 1	Voltage			Capacitance	DC leakage (max)			Impedance (max)	Case size
	Rated (85° C)	Derated (125° C)	Surge (85° C)		25° C	85° C	125° C		
	volts, dc	volts, dc	volts, dc	µF	µA	µA	µA	ohms	
CL7-BE200UNE	15	10	17.2	20	5	24	16	185	E1
CL7-BE700UNE	15	10	17.2	70	15	84	56	55	E2
CL7-BE201UNE	15	10	17.2	200	40	240	160	18.5	E3
CL7-BE411UNE	15	10	17.2	410	75	500	330	8.9	E4
CL7-BE591UNE	15	10	17.2	590	118	710	470	6.2	E5
CL7-BG100UNE	25	15	28.8	10	4	20	12	370	E1
CL7-BG350UNE	25	15	28.8	35	12	70	43	106	E2
CL7-BG101UNE	25	15	28.8	100	30	200	120	37	E3
CL7-BG211UNE	25	15	28.8	210	40	400	250	17.5	E4
CL7-BG301UNE	25	15	28.8	300	100	600	360	12.1	E5
CL7-BH080UNE	30	20	34.5	8	3	19	12	455	E1
CL7-BH290UNE	30	20	34.5	29	11	60	40	137	E2
CL7-BH800UNE	30	20	34.5	80	30	180	120	45.5	E3
CL7-BH171UNE	30	20	34.5	170	65	390	260	22	E4
CL7-BH241UNE	30	20	34.5	240	90	550	370	7.8	E5
CL7-BJ3R5TNE	50	30	57.5	3.5	3	14	9	1060	E1
CL7-BJ130TNE	50	30	57.5	13	10	52	31	285	E2
CL7-BJ360TNE	50	30	57.5	36	24	145	86	104	E3
CL7-BJ750TNE	50	30	57.5	75	45	300	180	49	E4
CL7-BJ111TNE	50	30	57.5	110	75	440	260	34	E5
CL7-BL020TNE	75	50	86.2	2	2	12	8	1850	E1
CL7-BL070TNE	75	50	86.2	7	8	42	28	550	E2
CL7-BL200TNE	75	50	86.2	20	23	120	80	185	E3
CL7-BL430TNE	75	50	86.2	43	45	260	170	86	E4
CL7-BL600TNE	75	50	86.2	60	60	360	240	62	E5
CL7-BN1R4SNE	100	65	115.0	1.4	2	11	7	2700	E1
CL7-BN050SNE	100	65	115.0	5	7	38	25	750	E2
CL7-BN140SNE	100	65	115.0	14	15	110	68	268	E3
CL7-BN290SNE	100	65	115.0	29	35	220	145	137	E4
CL7-BN410SNE	100	65	115.0	41	50	320	210	91	E5
CL7-BQR75SNE	150	100	172.0	0.75	2	9	6	5000	E1
CL7-BQ2R7SNE	150	100	172.0	2.7	6	30	20	1380	E2
CL7-BQ7R5SNE	150	100	172.0	7.5	15	84	56	490	E3
CL7-BQ150SNE	150	100	172.0	15	30	180	120	250	E4
CL7-BQ220SNE	150	100	172.0	22	45	270	175	170	E5

1 Complete type designation will include an additional symbol to indicate style.

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TABLE II. Cross-reference data.

FROM Type designation 1/ MIL-C-3965/23B	TO Part number 2/ MIL-C-39006/23	FROM Type designation 1/ MIL-C-3965 23B	TO Part number 2/ MIL-C-39006/23
	M39006/23-		M39006/23-
CL7-BE200UNE	1000	CL7-BJ750TNE	1018
CL7-BE700UNE	1001	CL7-BJ111TNE	1019
CL7-BE201UNE	1002	CL7-BL020TNE	1020
CL7-BE411UNE	1003	CL7-BL070TNE	1021
CL7-BE591UNE	1004	CL7-BL200TNE	1022
CL7-BG100UNE	1005	CL7-BL430TNE	1023
CL7-BG350UNE	1006	CL7-BL600TNE	1024
CL7-BG101UNE	1007	CL7-BN1R4SNE	1025
CL7-BG211UNE	1008	CL7-BN050SNE	1026
CL7-BG301UNE	1009	CL7-BN140SNE	1027
CL7-BH080UNE	1010	CL7-BN290SNE	1028
CL7-BH290UNE	1011	CL7-BN410SNE	1029
CL7-BH800UNE	1012	CL7-BQR75SNE	1030
CL7-BH171UNE	1013	CL7-BQ2R7SNE	1031
CL7-BH241UNE	1014	CL7-BQ7R5SNE	1032
CL7-BJ3R5TNE	1015	CL7-BQ150SNE	1033
CL7-BJ130TNE	1016	CL7-BQ220SNE	1034
CL7-BJ360TNE	1017		

1/ Complete type designation will include an additional symbol to indicate style.
2/ Part number referenced will replace styles CL72 and CL73.

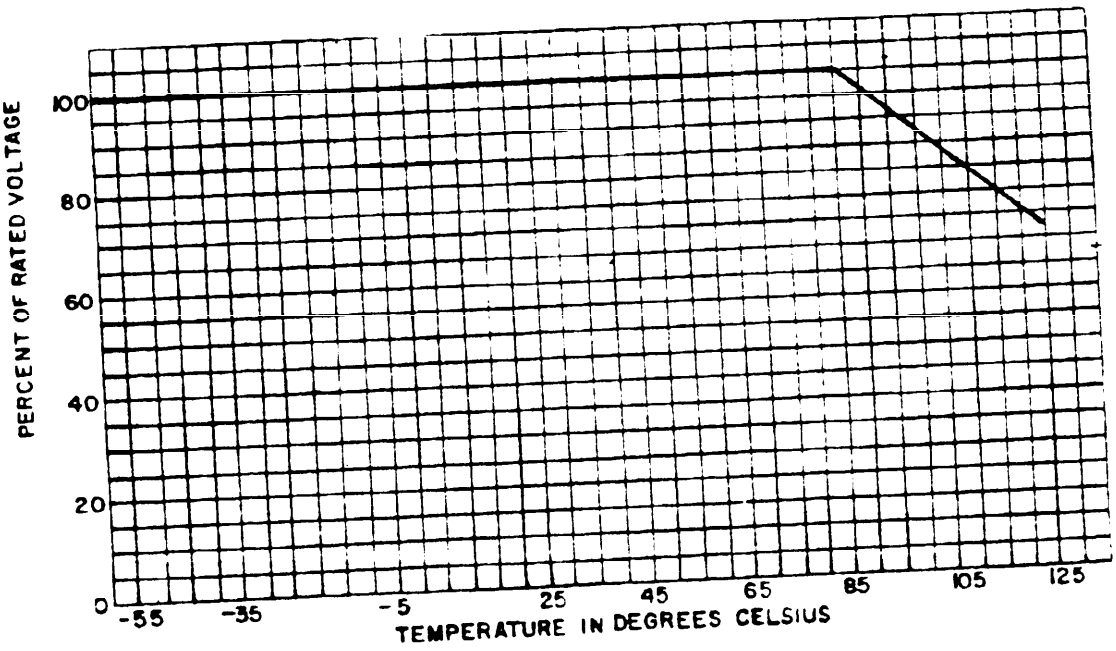


FIGURE 2. Voltage derating with temperature.

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REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1.
Case type - Tubular; uninsulated (CL72), insulated (CL73).
Seal type - Nonhermetic (symbol E).
Terminals - Axial wire-lead.
Rated temperature - -55° C to +85° C, voltage derated to +125° C (see figure 2).
Rated voltage - See table I.
Capacitance (cap) (nom) - See table I.
Capacitance tolerance - Symbol S (+30, -15%), T (+50, -15%), or U (+75, -15%) as shown in table I.

DC leakage (DCL) (max) (at +25° C):
See table I.

Dissipation factor (DF) (max):
20% (≤ 50 v) or 15% (> 50 v).

Seal: Not applicable.

Dielectric withstanding voltage (DWV):
Sleeving to case (min) - 2, 000 vdc.
Terminal to case - Not applicable.

Insulation resistance:
Sleeving to case (min): 100 megohms.
Terminal to case - Not applicable.

Temperature cycling:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - 115% of table I value.

Solderability: In accordance with MIL-C-3965.

Terminal strength:
Pull test - In accordance with MIL-C-3965.
Bend test - Not applicable.
Torque test - Not applicable.
Wire-lead bend test - In accordance with MIL-C-3965.
Twist test - Not applicable.

Surge voltage: See table I.

Moisture resistance:
DCL (max) - 125% of 25° C value of table I.
ΔCap - Within ±0% of initial measurement.
DF (max) - 115% of table I value.

Low temperature (storage):
DCL (max) - See table I (25° C).
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

1/ 125° C derated voltage referenced.

Stability at low and high temperature:
Step 1 (+25° C):
DCL (max) - See table I.
Cap - Within tolerance of table I value.
DV (max) - Shall not exceed initial requirement.
Step 2 (-55° C):
Impedance - See table I.
ΔCap (max) - -40% of step 1 value.
Step 3 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±5% of step 1 value.
DF (max) - Shall not exceed initial requirement.
Step 4 (+85° C):
DCL (max) - See table I.
ΔCap (Max) - +50% of step 1 value.
DF (max) - 55% (≤50 v) or 45% (>50 v).
1/ Step 5 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±10% of initial measurement.
DF (max) - 55% (≤30 v) or 45% (>30 v).
Step 6 (+25° C):
DCL (max) - See table I.
ΔCap - Within ±5% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Reverse voltage: Not applicable.

Life at +85° C:
2, 000-hr (qualification):
DCL (max) - See table I (25° C).
ΔCap - Within ±15% of initial measurement.
DF (max) - Shall not exceed initial requirement.
240-hr (group B):
DCL (max) - See table I (25° C).
ΔCap - Within ±10% of initial measurement.
DF (max) - Shall not exceed initial requirement.
1, 760-hr (group C):
DCL (max) - See table I (25° C).
ΔCap - Within ±15% of initial measurement.
DF (max) - Shall not exceed initial requirement.

Barometric pressure:
Test voltage - Rated voltage.

Life at +125° C:
DCL (max) - See table I (25° C).
ΔCap - Within ±15% of initial measurement.
DF (max) - Shall not exceed initial requirement.
DWV (min) (CL73, only) - 2000 vdc.
IR (min) (CL73, only) - 100 megohms.

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Voltage group (85° C rated voltage):
I - - - - - - - - - - 15 to 150 v, incl

NOTE: The capacitors covered by this specification sheet are substitutable for those covered by MIL-C-3965/23A. See table II.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL, MU
Navy - EC, SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Air Force - 80

Agent:
DSA - ES

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